Hello World

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Abstract

Here we will put together an even simpler document. We begin by adding no packages and then carefully introduce them compiling a little at a time.

1 First Section

Hello World!!!

2 Second Section

2.1 A Subsection

Here is a subsection!!!

2.2 Another Subsection

Here is another subsection!!!

3 Let's Try some Math

Here is an inline equation: $x^2 + 3x + 7 = 10$. Here is the same equation displayed:

$$x^2 + 3x + 7 = 10.$$

Then we can use a stacked equation to show solving this:

$$\begin{aligned} x^2 + 3x + 7 = 10 &\Rightarrow x^2 + 3x = 3 && (1)\\ &\Rightarrow x^2 + 3x + \frac{9}{4} = 5\frac{1}{4} && (2)\\ &\Rightarrow \left(x + \frac{3}{2}\right)^2 = \pm\sqrt{5\frac{1}{4}} && (3)\\ &\Rightarrow x = -\frac{3}{2} \pm \sqrt{5\frac{1}{4}}\,. && (4)\end{aligned}$$

What if we use special symbols like $\mathbb{N}$ or $\mathfrak{R}$? Yes, both of those worked well. What about $\mathcal{P}$ and ²⁄₃ and ✗? Yep, that worked too.

4 Let's Add a Diagram

Let us add in `\usepackage{tikz}` and make a diagram. Before even making a diagram we note that the compile time increases significantly while `latexml` parses what to do with `tikz`. Now here is a small figure:

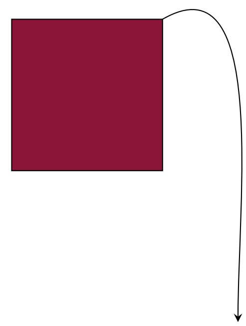

Figure 1: Our First Diagram

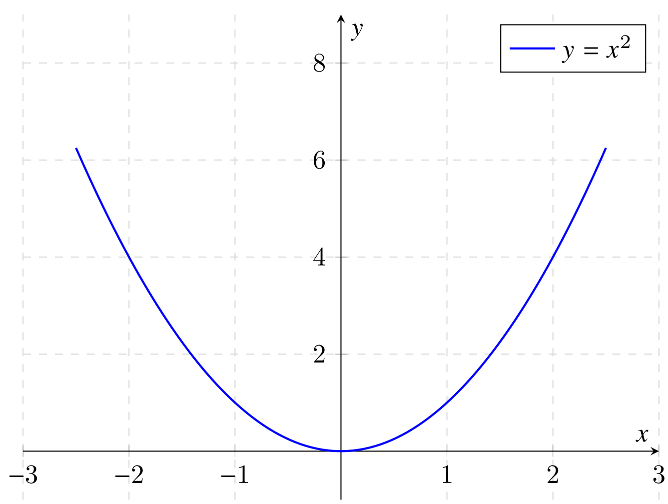

Figure 2: Parabola plot using PGFPlots. (from Gemini)

5 Now an Image

Figure 3: A Zombie

6 Miscellaneous Tests

This is another test
then clause

Table 1: The First 25 Prime Numbers in Alternating Colors

2	3	5	7	11
13	17	19	23	29
31	37	41	43	47
53	59	61	67	71
73	79	83	89	97

(a) Zombie One (b) Zombie Two

Figure 4: Two Zombies

This is in landscape in the PDF, the HTML understandably ignores this.

This is a test. This should split the text into multiple columns. It may be hard to see this in action without sufficient text. This is why the lipsum package is helpful. It looks like the HTML produced by LatexML also ignores the multicols environment, this also makes since.

test	row 5
test	row 5
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test	row 5
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test	row 5
test	row 5
test	row 5
test	row 5

Table 2: Example from colortbl manual

I should check that cross references work right like for figures 1 and 3 and tables 1 and 2.

```
lualatex Hello_World
latexml Hello_World.tex --dest=Hello_Latexml.xml
latexmlpost Hello_latexml.xml --dest=Hello_latexml.html --
    javascript="https://cdn.jsdelivr.net/npm/mathjax@3/es5/
    tex-mml-chtml.js?config=MML_HTMLorMML" --split --splitat
    =section --navigationtoc=context --css=andy-navbar.css
    --css=normalize.css --urlstyle=file --timestamp=0 --
    javascript=andy-navbar.js
```

7 Home Built or Defined Macros

♥ ♦ ✓, ✗

8 Notes:

LatexML does not seem to like the document meta data. This supports the idea that there should be two shells, one with metadata and one without. The only problem with this is that the alt text in tikz seems to only be recognized when the metadata is present and compiling with lualatex. This supports the view that it may be best practices to create diagrams separately and then import them. Within the document the `\iflatexml` command seems to work fine as long as the compiler can find `latexml.sty`. So far following the instructions at this site `https://hackmd.io/@UoL-IWG/latexml` has LatexML working like a champ; the question is about what packages this can handle. There is also important information in the GitHub repository for the LaTeXML project: `https://github.com/brucemiller/LaTeXML`. The package minted continues to be an issue, it is not supported natively by LaTeXML or tex4ht and is not listed as compatible with the work from the LaTeX Tagging project; we need to learn fancyvrb. Tried fancyvrb and listings and listings seems to be the better of the two.